

LAB EXP:1

python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/ANLNA/One

Training data:

Sky	AirTemp	Humidity	Wind	Water	Forecast	Target
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

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FIND-S TRACE

Initial hypothesis h0 = ['0', '0', '0', '0', '0', '0']

Example 1: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same'] -> POSITIVE

h1 = ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

Example 2: ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same'] -> POSITIVE

h2 = ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Example 3: ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change'] -> NEGATIVE (ignored)

h3 = ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Example 4: ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change'] -> POSITIVE

h4 = ['Sunny', 'Warm', '?', 'Strong', '?', '?']

=====

MOST SPECIFIC HYPOTHESIS (FIND-S output)

Sky : Sunny
AirTemp : Warm
Humidity : ?
Wind : Strong
Water : ?
Forecast : ?
h = ['Sunny', 'Warm', '?', 'Strong', '?', '?']

Classification demo:

['Sunny', 'Warm', 'Normal', 'Strong', 'Cool', 'Change'] -> Yes

['Rainy', 'Cold', 'Normal', 'Light', 'Warm', 'Same'] -> No

LAB EXPERIMENT - 1.py - C:/Users/ANLNA/OneDrive/MLA0203 ML/LAB EXPERIMENT - 1.py (3.14.2)

```
import csv
import sys

ATTRIBUTES = ["Sky", "AirTemp", "Humidity", "Wind", "Water", "Forecast"]

DATA = [
    ["Sunny", "Warm", "Normal", "Strong", "Warm", "Same", "Yes"],
    ["Sunny", "Warm", "High", "Strong", "Warm", "Same", "Yes"],
    ["Rainy", "Cold", "High", "Strong", "Warm", "Change", "No"],
    ["Sunny", "Warm", "High", "Strong", "Cool", "Change", "Yes"],
]

def load_csv(path):
    with open(path, newline='') as f:
        rows = [[c.strip() for c in r] for r in csv.reader(f) if r]
    return rows[0][:-1], rows[1:]

def find_s(attributes, data, verbose=True):
    n = len(attributes)
    h = ["0"] * n

    if verbose:
        print("=" * 64)
        print("FIND-S TRACE")
        print("=" * 64)
        print(f"Initial hypothesis h0 = {h}\n")

    for i, row in enumerate(data, start=1):
        x, label = row[0:-1], row[-1].strip().lower()

        if label != "yes":
            if verbose:
                print(f"Example {i}: {x} -> NEGATIVE (ignored)")
                print(f"      h{i} = {h}\n")
            continue

        if verbose:
            print(f"Example {i}: {x} -> POSITIVE")
```

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LAB EXP:2

```
import csv

f = open("training_data.csv", "w")
f.write("""Sky,AirTemp,Humidity,Wind,Water,Forecast,EnjoySport
Sunny,Warm,Normal,Strong,Warm,Same,Yes
Sunny,Warm,High,Strong,Warm,Same,Yes
Rainy,Cold,High,Strong,Warm,Change,No
Sunny,Warm,High,Strong,Cool,Change,Yes
""")
f.close()

data = list(csv.reader(open("training_data.csv")))
header = data[0]
data = data[1:]
n = len(header) - 1

values = []
for i in range(n):
    values.append(sorted(set(row[i] for row in data)))

def covers(h, x):
    for i in range(n):
        if h[i] != "?" and h[i] != x[i]:
            return False
    return True

def more_general(g, s):
    for i in range(n):
        if g[i] != "?" and g[i] != s[i] and s[i] != "0":
            return False
    return True

S = ["0"] * n
G = ["?" * n]
print("Initial S:", S)
print("Initial G:", G)

for x, row in enumerate(data, 1):
    x = row[0:-1]
    print(f"Example {x}: {x}")
    if row[-1] == "yes":
        G = [g for g in G if covers(g, x)]
        if S[0] == "0":
            S = x
        else:
            for i in range(n):
                if S[i] != x[i]:
                    S[i] = "?"
    else:
        newG = []
        for g in G:
            if not covers(g, x):
                newG.append(g)
        for i in range(n):
```

```
===== RESTART: C:/Users/ANLNA/AppData/Local/Programs/Python/Python314/LAB 2 ML.py =
Initial S: ['0', '0', '0', '0', '0', '0', '0']
Initial G: [['?', '?', '?', '?', '?', '?', '?']]

Example 1 ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same'] Yes
S: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
G: [['?', '?', '?', '?', '?', '?', '?']]

Example 2 ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same'] Yes
S: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
G: [['?', '?', '?', '?', '?', '?', '?']]

Example 3 ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change'] No
S: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
G: [['Sunny', '?', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?', 'Same']]

Example 4 ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change'] Yes
S: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
G: [['Sunny', '?', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?', '?']]

Final S: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
Final G: [['Sunny', '?', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?', '?']]

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```